

SMART CITY HACKATHON · MAGDEBURG

MAGmagdeburg

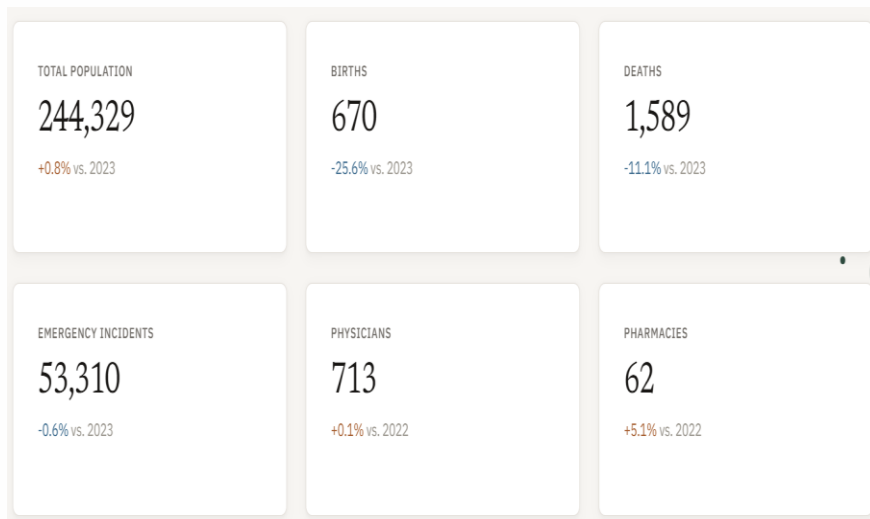
Making Healthcare Access Visible

An Open Data Smart City Dashboard for Magdeburg — using healthcare, demographic, and emergency service data to identify underserved neighborhoods before they become critical.



THE PROBLEM

Magdeburg Is Growing — But the Story Is Complex



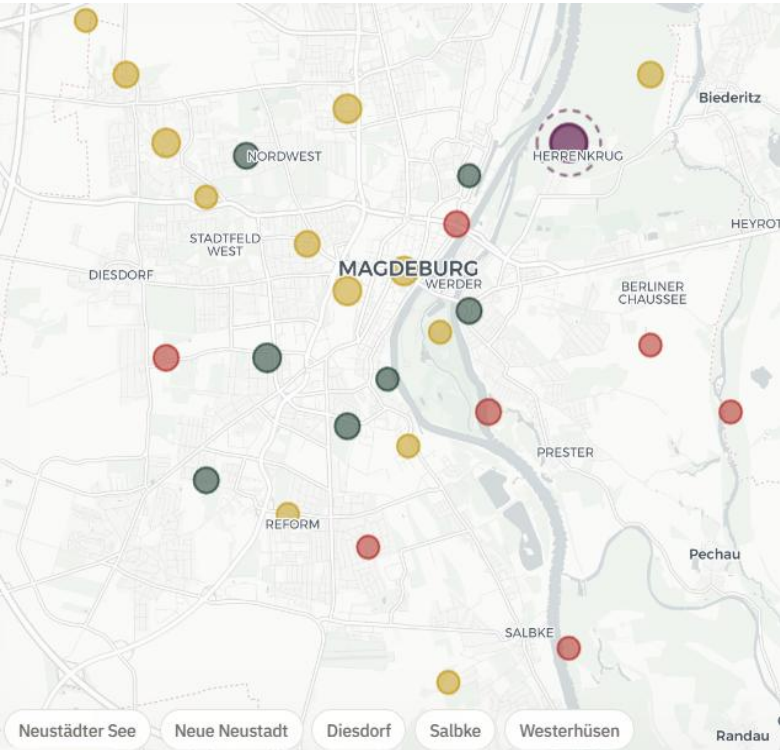
What the Data Reveals

- Population grew by **+12,000** since 2010
- Deaths **exceed births** every year
- Growth sustained mainly through **migration**
- Population is **aging** — healthcare demand rising

 **Key Question:** Are healthcare services available where people need them most?

THE PROBLEM

Healthcare Access Is Not Equal Across Districts



53,000+

Emergency incidents in 2024

Aging Population

Growing elderly demand across districts

Uneven Distribution

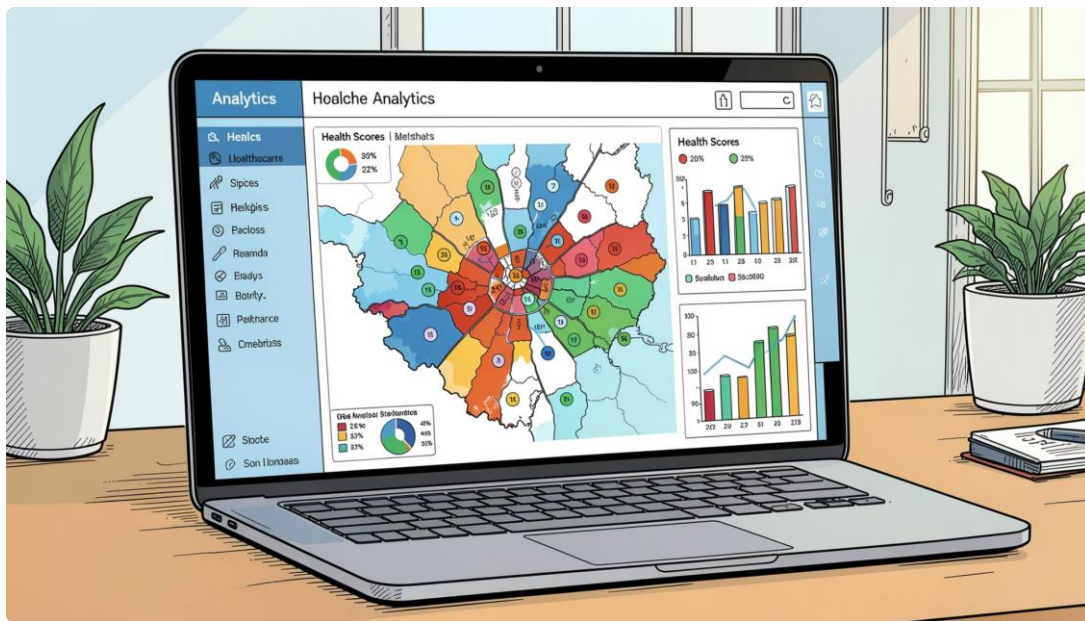
Physicians and pharmacies concentrated in select areas

Hidden Gaps

No tool exists to identify vulnerable neighborhoods

For elderly residents, proximity to healthcare directly affects quality of life — yet no one can easily see which districts are most at risk.

MAGmagdeburg Healthcare Intelligence Dashboard



A District-Level Decision Intelligence Platform

Built entirely using public Magdeburg open datasets
– transforming raw data into actionable healthcare insights.

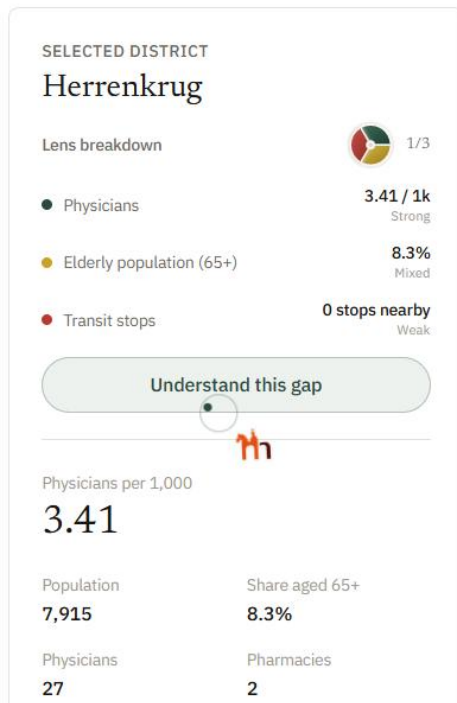
→ Citizens & Families
Find healthcare-friendly neighborhoods

→ Healthcare Planners
Target investments where needed most

→ City Administration
Make evidence-based decisions

Triple Lens Healthcare Analysis

Most dashboards show numbers. MAGmagdeburg shows **where action is needed** – combining three critical dimensions into a single **Compound Access Index**.



Lens 1 · Physician Density

Are enough doctors available per 1,000 residents?

Lens 2 · Elderly Population

Is the population aging faster than average?

Lens 3 · Transit Stops

Is the public transport accessible?

From Raw Data to Actionable Insights

Each district receives a composite score across all three lenses, classified into four clear action tiers.



Strong Access

Good performance across all three indicators



Moderate Concern

One or two indicators showing strain



High Concern

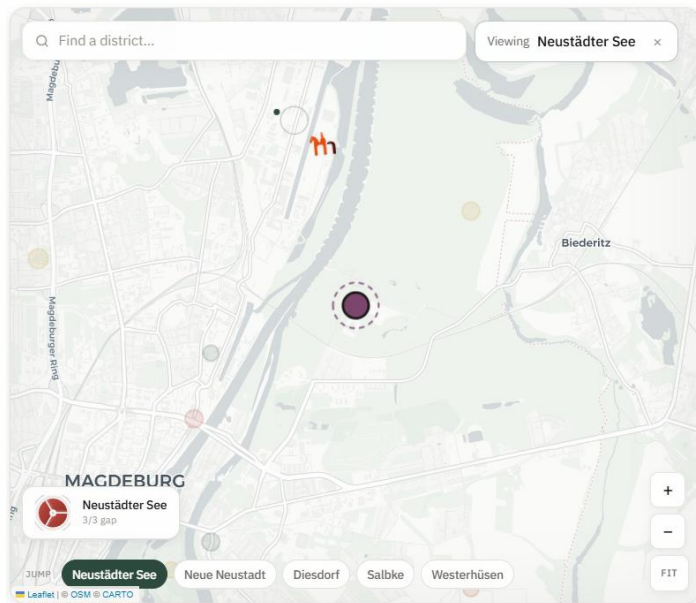
Multiple indicators in critical range



Triple Gap District

Weak performance in all three lenses simultaneously –
priority for intervention

We Found Healthcare Gaps Hidden in City Data



SELECTED DISTRICT Neustädter See

Triple gap

Lens breakdown

Physicians



3/3

0.78 / 1k
Weak

Elderly population (65+)

23.3%

Weak

Transit stops

5 stops nearby

Weak

Understand this gap

Physicians per 1,000

0.78

Population

1,290

Share aged 65+

23.3%

Physicians

1

Pharmacies

0

Care priority

High priority - #6

above-average aging, low physician density

Case Study: Neustädter See

Lower Physician Availability

Fewer doctors per capita than city average

Higher Elderly Population

Above-average demographic risk score

Greater Healthcare Demand

Elevated emergency service call volume



Result: Identified as a **Triple Gap District** – a priority area for future healthcare investment.

IMPACT

Turning Data Into Decisions



Citizens & Families

Identify healthcare-friendly neighborhoods and support elderly relatives with informed choices.



Healthcare Planners

Target investments precisely — knowing which districts need physicians, clinics, or new facilities.



City Administration

Replace guesswork with evidence — distribute resources where they are needed most.

WHY IT WINS

Open Data → Actionable Intelligence



Added Value

Directly addresses a real, unmet healthcare access gap in Magdeburg



Creativity

Triple Lens methodology creates a novel Compound Access Index



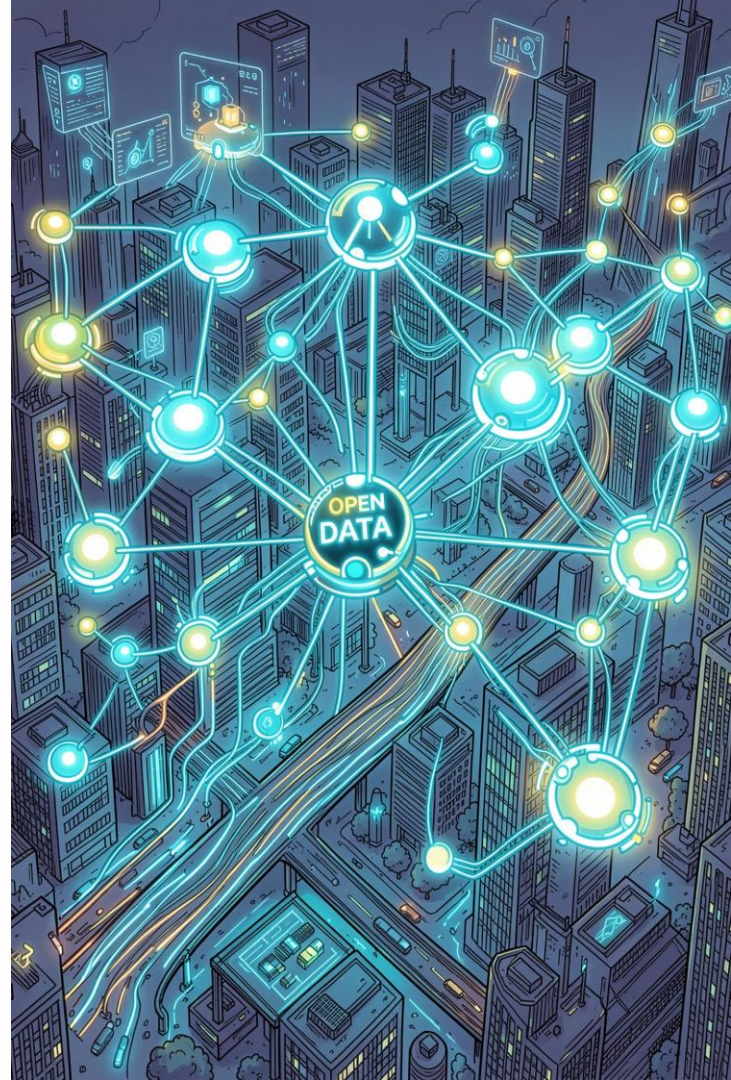
Reusability

Built entirely on public datasets — no proprietary data required



Scalability

Framework replicable in any city with population and health data





OUR VISION

A Smarter and Healthier Magdeburg

"You cannot improve what you cannot see. MAGmagdeburg makes healthcare access visible."

Many dashboards show data. MAGmagdeburg shows **where action is needed** — helping citizens, planners, and decision-makers build a healthier future for every district.

Identify

Underserved districts before they
become critical

Invest

Smarter, targeted healthcare
infrastructure

Improve

Quality of life for every Magdeburg citizen